(i) Capacitive reactance,

$$X_C = \frac{1}{\omega C} = \frac{1}{2\pi f C}$$

$$= \frac{1}{2 \times \pi \times 50 \times \frac{10^{-4}}{\pi}} = 100\,\Omega$$

(ii) RMS value of current,

$$I_{RMS} = \frac{V_{RMS}}{X_C} = \frac{220}{100} = 2.2\,\text{A}$$

(iii) $V_m = 220 \times \sqrt{2} = 311\ \text{V}$

$$I_m = 2.2 \times \sqrt{2} = 3.1\text{A}$$

Therefore,

$$\upsilon = 311 \sin 314t$$

$$i = 3.1 \sin\left(314t + \pi/2\right)$$

4.7.6 AC circuit containing a resistor, an inductor and a capacitor in series – Series RLC circuit

Consider a circuit containing a resistor of resistance R, an inductor of inductance L and a capacitor of capacitance C connected across an alternating voltage source (Figure 4.46). The instantaneous value of the alternating voltage is given by

$$\upsilon = V_m \sin \omega t$$

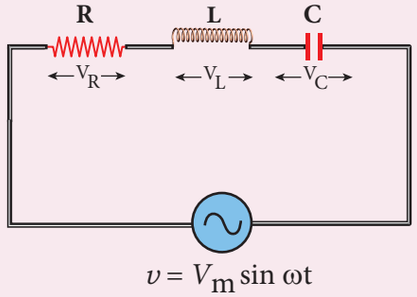


Figure 4.46 AC circuit containing R, L and C

Let i be the resulting current in the circuit at that instant. As a result, the voltage is developed across R, L and C.

We know that voltage across R (V_R) is in phase with i, voltage across L (V_L) leads i by $\pi/2$ and voltage across C (V_C) lags behind i by $\pi/2$.

The phasor diagram is drawn with current as the reference phasor. The current is represented by the phasor $\overrightarrow{OI}$, V_R by $\overrightarrow{OA}$; V_L by $\overrightarrow{OB}$ and V_C by $\overrightarrow{OC}$ as shown in Figure 4.47.

The length of these phasors are

$$OI = I_m,\ OA = I_m R,\ OB = I_m X_L;\ OC = I_m X_C$$

The circuit is either effectively inductive or capacitive or resistive depending on the value of V_L or V_C. Let us assume that $V_L > V_C$. Therefore, net voltage drop across L-C combination is $V_L - V_C$ which is represented by a phasor $\overrightarrow{OD}$.

By parallelogram law, the diagonal $\overrightarrow{OE}$ gives the resultant voltage υ of V_R and $(V_L - V_C)$ and its length OE is equal to V_m. Therefore,

$$V_m^2 = V_R^2 + \left(V_L - V_C\right)^2$$

$$V_m = \sqrt{\left(I_m R\right)^2 + \left(I_m X_L - I_m X_C\right)^2}$$

$$= I_m \sqrt{R^2 + \left(X_L - X_C\right)^2} \quad \text{or}$$

$$I_m = \frac{V_m}{\sqrt{R^2 + \left(X_L - X_C\right)^2}} \quad \text{or} \qquad (4.46)$$

$$I_m = \frac{V_m}{Z}$$

where $Z = \sqrt{R^2 + \left(X_L - X_C\right)^2}$ (4.47)

Z is called impedance of the circuit which refers to the effective opposition to the current by the series RLC circuit.

UNIT-4(XII-Physics_Vol-1).indd 244 17-02-2020 13:19:41